

RESEARCH ARTICLE

Functional insight into East Asian-specific genetic risk loci for Alzheimer's disease

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Abstract

INTRODUCTION: The functional study of genetic risk factors for Alzheimer's disease (AD) provides insights into the underlying mechanisms and identification of potential therapeutic targets. Investigating AD-associated genetic loci identified in East Asian populations using single-nucleus RNA-sequencing data may identify novel functional genetic contributors.

METHODS: Cell type-specific expression quantitative trait loci (eQTL) and peak-to-gene links were used to identify functional genes associated with 26 genetic loci from seven genome-wide association studies (GWAS) for AD in East Asians.

RESULTS: *KCNJ6* and *MAPK1IP1L* were identified as significant eQTLs with AD risk loci. AD risk loci were in peaks related to four genes, with *CLIC4* being connected across different cell types. Genes identified in European and East Asian GWAS interacted within networks and were enriched in AD pathology pathways in astrocytes.

DISCUSSION: Our findings suggest *KCNJ6* and *CLIC4* as novel AD-associated functional genes, providing insight into the genetic architecture of AD in East Asians.

KEYWORDS

Alzheimer's disease, cell type-specific analysis, diversity population, East Asian, genome-wide association analysis

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Highlights

- Integrated functional analysis of Alzheimer's disease (AD) loci in seven East Asian genome-wide association studies (GWAS) was performed.
- Cell type-specific expression quantitative trait loci (eQTLs) and assay for transposase-accessible chromatin peaks were used to identify AD functional genes.
- An AD risk variant was linked to *KCNJ6* through an oligodendrocyte progenitor cell-specific eQTL.
- An AD risk variant maps to open chromatin, linked to *CLIC4* across six cell types.
- Astrocyte differentially expressed genes by AD pathology are enriched in East Asian and European GWAS genes.

1 | BACKGROUND

Alzheimer's disease (AD) is known as the leading cause of dementia, affecting > 57 million individuals worldwide.¹ With an estimated heritability of 60% to 80%, genetic studies are crucial for understanding AD mechanisms, prompting genome-wide association studies (GWAS) to uncover genetic factors.² One of the largest AD GWAS studies, involving a cohort of ≈ 1 million individuals, identified 75 genetic risk loci associated with AD.³ However, most GWAS studies have primarily focused on European populations, potentially limiting the discovery of genetic factors unique to other ethnic groups and highlighting the importance of conducting genetic studies across diverse populations to uncover novel genetic variants.⁴

GWAS conducted with non-European populations have identified novel risk loci that were not previously identified in European studies. A meta-analysis of AD GWAS, including European, East Asian, African American, and Caribbean Hispanic populations, uncovered two novel loci.⁵ Additionally, this study revealed that 37% of the leading single nucleotide polymorphisms (SNPs) from European GWAS showed significant heterogeneity. Furthermore, a multi-ethnic genetic study of amyloid beta ($A\beta$) using amyloid positron emission tomography imaging confirmed differential effects of apolipoprotein E $\epsilon 4$ and $\epsilon 2$ across diverse populations.⁶ Overall, these findings highlight the necessity of conducting research on diverse populations in genetic studies of AD, emphasizing the presence of distinct genetic profiles across populations.

Recently, GWAS studies have focused on East Asian populations, particularly those in South Korea, China, and Japan.⁷ These studies have identified novel risk loci, including variants with high minor allele frequencies (MAFs) specific to East Asians. A variant near *CACNA1A* identified in Korean GWAS shows a higher MAF in East Asians compared to Europeans.⁸ Additionally, rare variants in the coding regions of the *SHARPN* gene show significant associations in Japanese and Korean populations, due to their relatively high MAFs in East Asian populations.^{9–11} These findings emphasize that due to differences in MAFs among ethnicities, signals that were challenging to detect in European studies could be identified in East Asian studies.

However, most East Asian GWAS studies have not leveraged diverse multi-omics data, which is crucial for linking risk loci to specific genes and uncovering their functional roles.^{12,13} Furthermore, due to the cell heterogeneity of the brain tissue, cell type-specific analysis is crucial for a deeper understanding of the underlying mechanism of AD. Single-nucleus transcriptomics (snRNA) and single-nucleus assay for transposase-accessible chromatin (snATAC) enable the identification of cell type-specific alterations in transcriptomes and epigenomes.^{14,15} Therefore, these analyses offer insights into the genes and mechanisms linked to risk loci identified in East Asian populations at a cell type-specific level.

In this study, we aimed to explore genes related to AD risk loci identified in previous East Asian GWAS using cell type-specific multi-omics data. Specifically, our approach involves leveraging cell type-specific expression quantitative trait loci (eQTL) and peak-to-gene links to identify relevant genes with the risk loci. Additionally, we assessed how the expression of these genes changed across different cell types and AD pathology. Furthermore, we conducted a comparison between genes identified in European and East Asian GWAS to elucidate the relationship among identified genes across different ethnicities. Through this approach, we aimed to investigate East Asian risk loci at the single-cell level and elucidate the relationships of genes identified across different ethnicities.

2 | METHODS

2.1 | Gene prioritization of AD susceptibility loci from East Asian GWAS studies

We conducted gene prioritization using AD susceptibility loci identified in seven East Asian GWAS studies,^{8,16–21} as summarized in a previous review paper.⁷ These studies focused on Japanese, Chinese, and South Korean populations. Specifically, 5 loci were identified in the Japanese population, 8 loci in the Chinese population, and 13 loci in the South Korean population. For each lead SNP, gene prioritization was performed using cell type-specific eQTLs and peak-to-gene links.

Cell type-specific eQTLs derived from the snRNA-sequencing data of 424 individuals in the Religious Orders Study and Memory and Aging Project (ROSMAP) cohort were used.²² We used the brain cell type-specific eQTL portal (https://vmenon.shinyapps.io/rosmap_snRNAsequencing_eqtl/) to identify genes with significant eQTLs associated with the lead SNPs identified in East Asian populations.

For peak-to-gene links, we used p data from an integrative study of snRNA-sequencing and snATAC-sequencing involving 92 individuals from the ROSMAP cohort.¹⁵ The integration of snRNA-sequencing and snATAC-sequencing data identified chromatin-accessible peaks associated with genes. We extracted genes linked to peaks containing the lead SNPs identified in East Asian populations. Visualization was conducted using the University of California Santa Cruz Genome Browser.²³

2.2 | Functional analyses

For functional analysis, we used both previously identified genes from East Asian GWAS studies and newly prioritized genes using cell type-specific eQTLs and peak-to-gene links. From the previous review paper summarizing East Asian GWAS studies, 26 genes associated with common variants and two genes (*TREM2* and *SHARPIN*) identified through rare variants analyses were reported.^{7-11,16-21,24} Additionally, through gene prioritization, we incorporated four genes that were not identified in the previous studies (*KCNJ6*, *MAPK11P1L*, *AC016292.1*, and *AC100800.1*), resulting in a total of 32 genes for functional analyses.

2.3 | Cell type-specific differentially expressed gene analysis

Cell type-specific expression and gene trajectories were obtained from the Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD), accessed via the Allen Brain Map web application (https://sea-ad.shinyapps.io/ad_gene_trajectories/).²⁵ Specifically, scRNA-sequencing measured gene expression across ≈ 1.7 million cells from the medial temporal gyrus (MTG), while pseudo-progression scores were computed using machine learning methods based on quantitative staining of key pathological proteins in the MTG.

Cell type-specific differentially expressed genes (DEGs) were derived from a previous study of the ROSMAP cohort.¹⁴ Briefly, DEGs were calculated using snRNA-sequencing data across 427 individuals, covering 2.3 million cells with AD pathology-associated phenotypes. DEG analyses were performed using the R package muscat, aggregating single-cell measurements into pseudobulk data for each participant and cell type. The global AD burden was assessed through standardized counts of neuritic plaques, diffuse plaques, and neurofibrillary tangles. Cognitive decline resilience (CDR) scores were computed based on the measure estimating the rate of change of global cognitive function over time. False discovery rate correction was applied for statistical significance. We used DEG lists adjusted with DecontX and RUVr methods from the AD/Aging Brain Atlas (https://compbio.mit.edu/ad_aging_brain/). Heatmap visualization was performed using the ggplot2 package (version 3.5.0) in R (version 4.3.1).

RESEARCH IN CONTEXT

- Systematic review:** The authors conducted a review of the literature using traditional sources including PubMed and Google Scholar. Cell type-specific functional studies for Alzheimer's disease (AD) risk loci identified in East Asians using single-cell transcriptome and epigenome data have not been performed.
- Interpretation:** This study identifies *KCNJ6* and *MAPK11P1L* as significant expression quantitative trait loci signals for AD risk loci in East Asians. Peak-to-gene analysis links AD risk loci to four genes, including *CLIC4*. This study also shows that differentially expressed genes in astrocytes by AD pathology are enriched in East Asian and European genome-wide association study genes. These findings suggest *KCNJ6* and *CLIC4* as novel AD-associated functional genes.
- Future directions:** More studies using single-cell multi-omics data from large-scale East Asian cohorts should investigate the roles of *KCNJ6* and *CLIC4* in AD.

2.4 | Comparative analysis of genes identified in European and East Asian GWAS

Comparative analyses were conducted between genes identified in East Asian GWAS and those identified in European GWAS. We used 76 genes identified in a European GWAS study involving > 700,000 individuals.³ Network analysis using the STRING database (version 12.0)²⁶ was performed with 76 genes identified in European GWAS and 32 genes identified in the East Asian GWAS. Genes not connected within the network were excluded, and closely positioned genes within the network were grouped for pathway analyses using Enrichr.²⁷⁻²⁹ Significant pathways identified from Kyoto Encyclopedia of Genes and Genomes 2021 or Gene Ontology Biological Processes 2023, particularly those most relevant to AD, were annotated on the network.

Next, gene set enrichment analysis (GSEA) was performed to test for enrichment using the gene sets identified from the East Asian or European GWAS.^{30,31} For testing enrichment, the log fold changes of DEGs derived from the global AD pathology burden in the ROSMAP-Massachusetts Institute of Technology dataset were used as the input data.¹⁴

3 | RESULTS

3.1 | Prioritization of GWAS genes through cell type-specific eQTL analysis identified the *KCNJ6* gene

A previous review paper summarized seven East Asian GWAS studies, identifying a total of 26 AD risk loci.⁷ To investigate the regulatory

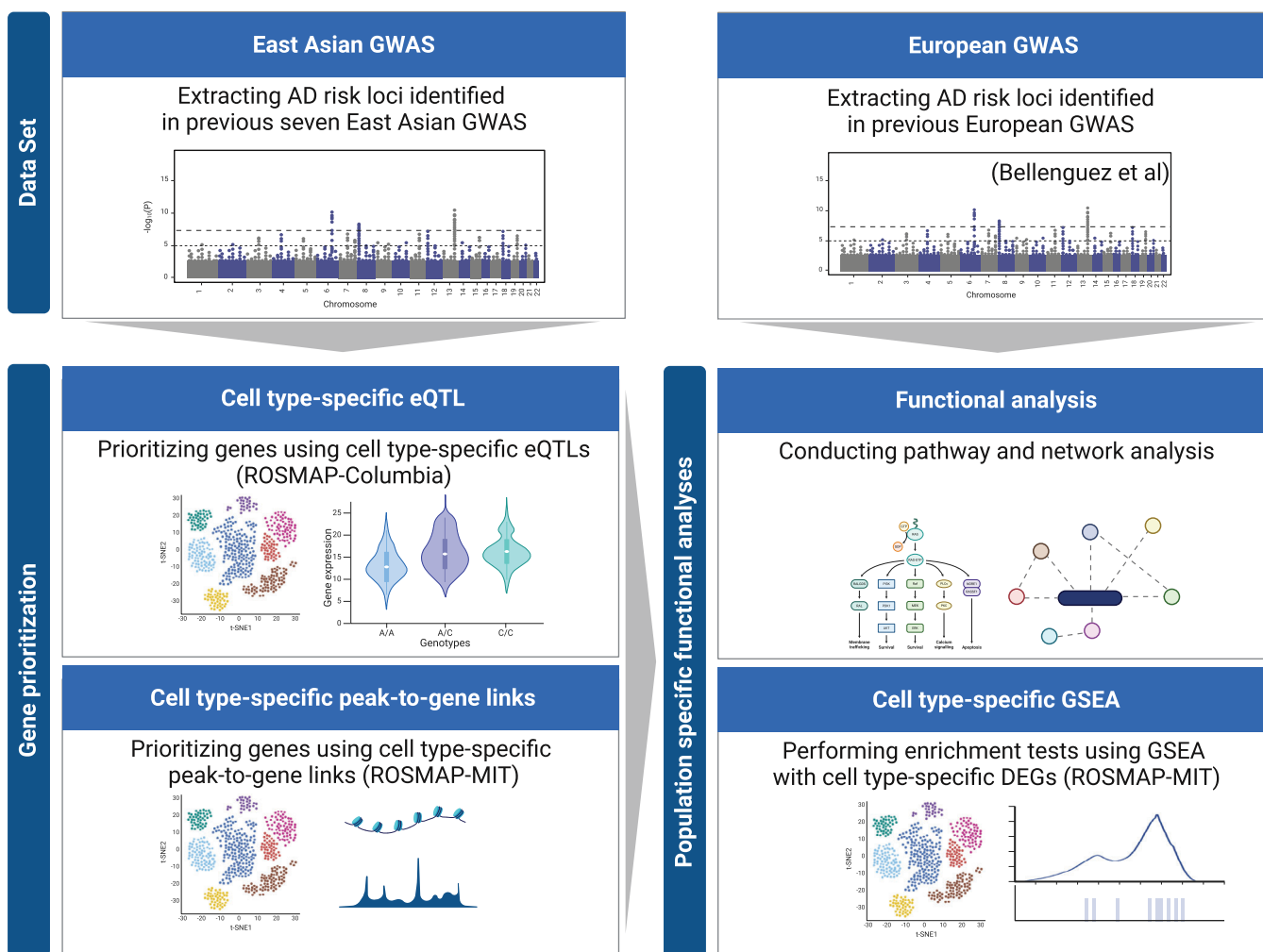


FIGURE 1 Study overview. AD, Alzheimer's disease; DEG, differentially expressed genes; eQTL, expression quantitative trait loci; GSEA, gene set enrichment analysis; GWAS, genome-wide association study; ROSMAP, Religious Orders Study Memory and Aging Project.

impact of the 26 lead variants on gene expression, we used cell type-specific eQTLs derived from snRNA sequencing data (Figure 1).²² In excitatory neurons, rs72713460 was significantly associated with expression levels of *MAPK1IP1L* ($p = 3.17 \times 10^{-6}$), while in oligodendrocyte progenitor cells (OPCs), rs928771 was significantly associated with expression levels of *KCNJ6* ($p = 5.57 \times 10^{-29}$; Figure 2A). These genes had not been identified in prior East Asian GWAS studies.

Next, we performed pathway and network analyses with previously reported genes and newly identified genes to identify the interactions of these genes. Pathway analysis with GO molecular function revealed that previously identified genes *SORL1*, *TREM2*, and *CACNA1A* were included in the $A\beta$ binding pathway (Figure 2B). Furthermore, *SORL1* and *TREM2* were part of the low-density lipoprotein particle binding pathway.

Network analysis demonstrated that *KCNJ6* interacts with *CACNA1A* and *KCNJ15*. Notably, *KCNJ6* and *KCNJ15* were identified in a GWAS in a Chinese population, whereas *CACNA1A* was identified in a GWAS in a South Korean population. These findings

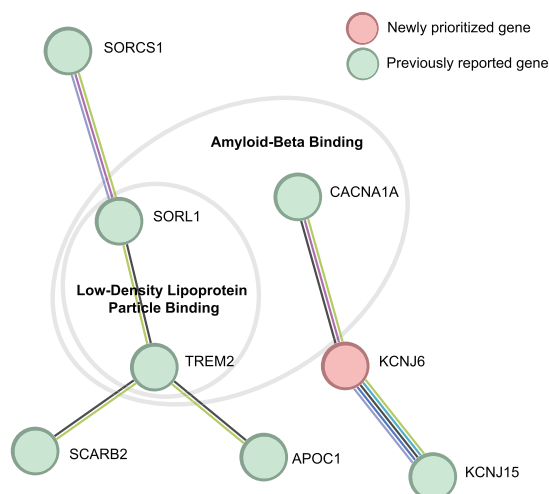
highlight that *KCNJ6*, a newly identified gene using cell type-specific eQTL analysis, was associated with previously reported genes in East Asian GWAS.

We further investigated the cell type-specific expression patterns of *KCNJ6*. Using the SEA-AD data, we observed that *KCNJ6* was highly expressed in neuronal cell types, particularly in the chandelier inhibitory neuron and the L6b excitatory neuron cell types (Figure 2C). Furthermore, *KCNJ6* expression was observed to increase with AD pseudo-progression across most cell types. Cell type-specific differential expression analysis using the ROSMAP differential expression dataset¹⁴ indicated that the *KCNJ6* gene was significantly differentially expressed in neurons, particularly in cognitive function phenotypes (Figure 2D). Individuals with a positive rate of change in cognitive performance showed lower *KCNJ6* expression in eight excitatory neuron subtypes and four inhibitory neuron subtypes. Overall, *KCNJ6* exhibits high expression in neuronal cell types, showing a pattern of increased expression with AD pseudo-progression or lower cognitive performance change rates.

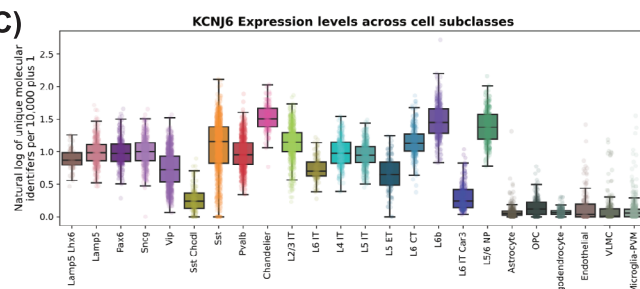
(A)

Cell types	Gene symbol	SNP rsID	Ref	Alt	Alt AF	beta	se	p-value
Excitatory neuron	MAPK1IP1L	rs72713460	G	T	0.191	0.146	0.0309	3.17E-06
Oligodendrocyte progenitor cells	KCNJ6	rs928771	T	G	0.5047	-0.64	0.0526	5.57E-29

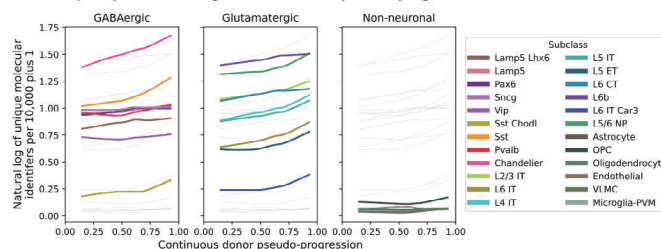
(B)



(C)



KCNJ6 Expression change across disease pseudo-progression - Subclass Resolution



(D)

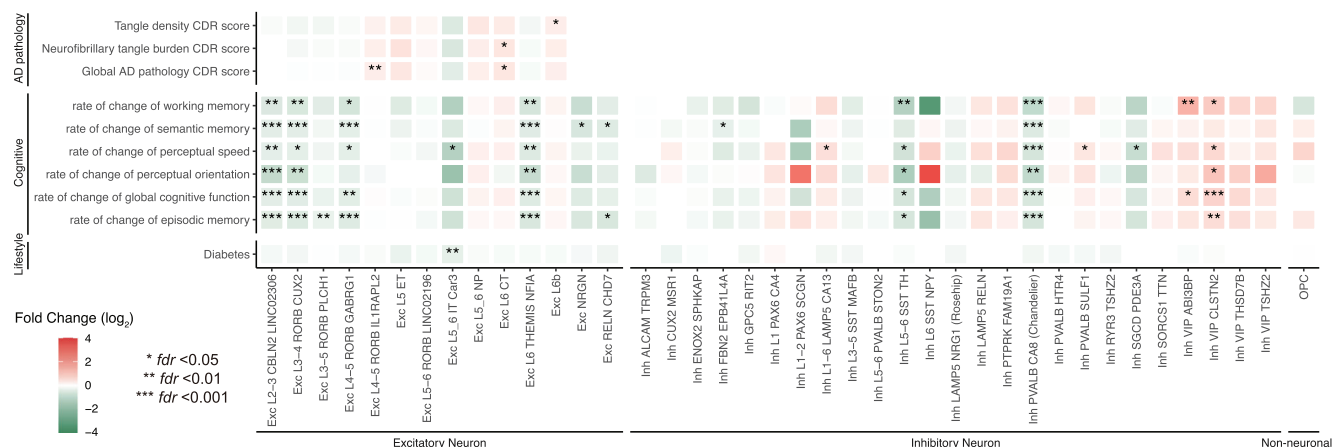


FIGURE 2 Functional and cell type-specific expression analysis of *KCNJ6*. A, Genes with significant cell type-specific eQTLs with lead variants identified in East Asian GWAS. B, Network and pathway analysis of newly identified genes through cell type-specific eQTLs and previously reported genes in East Asian GWAS. Green nodes represent previously reported genes, and red nodes represent newly prioritized genes. The different line colors represent interaction types: light blue for curated databases, pink for experimentally determined interactions, green for gene neighborhood, red for gene fusions, blue for gene co-occurrence, yellow-green for text mining, black for co-expression, and purple for protein homology. C, Cell type-specific expression and expression patterns across pseudo-progression of AD pathology of *KCNJ6*. Plotting was performed using SEA-AD gene expression trajectories viewer. D, Heatmap of cell type-specific DEGs for *KCNJ6*. Significance levels are indicated as * $fdr < 0.05$, ** $fdr < 0.01$, and *** $fdr < 0.001$. AD, Alzheimer's disease; AF, allele frequency; Alt, alternate allele; CDR, cognitive decline resilience; DEG, differentially expressed genes; fdr, false discovery rate; eQTL, expression quantitative trait loci; GWAS, genome-wide association study; Exc, excitatory neuron; Inh, inhibitory neuron; OPC, oligodendrocyte progenitor cells; Ref, reference allele; SEA-AD, Seattle Alzheimer's Disease Brain Cell Atlas; SNP, single nucleotide polymorphism.

3.2 | Characterizing cell type-specific regulatory regions of *CLIC4*

We examined whether AD risk loci identified in East Asian GWAS are located within chromatin accessible peaks linked with genes. Using

cell type-specific peak-to-gene links, we identified four lead variants located within peak regions linked to *CLIC4*, *LRIG1*, *AC016292.1*, and *AC100800.1* genes (Table 1). Particularly for *CLIC4*, peaks were observed in multiple cell types, and rs12063304, identified in a South Korean GWAS study, was located near the transcription start site (TSS)

TABLE 1 Genes linked with lead variants identified in East Asian GWAS through cell type-specific peak-to-gene links.

Peak	Gene	Cell types	Distance
chr1:24744913-24745413	<i>CLIC4</i>	OPC	−219
chr1:24744927-24745427	<i>CLIC4</i>	Astrocytes	−205
chr1:24744962-24745462	<i>CLIC4</i>	Microglia	−170
chr1:24744967-24745467	<i>CLIC4</i>	Vascular	−165
chr1:24744987-24745487	<i>CLIC4</i>	Inhibitory neurons	−145
chr1:24745141-24745641	<i>CLIC4</i>	Oligodendrocytes	9
chr8:18689650-18690150	<i>AC100800.1</i>	Oligodendrocytes	−31005
chr17:1179496-1179996	<i>AC016292.1</i>	Excitatory neurons	−2147
chr17:1179508-1180008	<i>AC016292.1</i>	Inhibitory neurons	−2135
chr3:66492030-66492530	<i>LRIG1</i>	Astrocytes	8983

Abbreviations: GWAS, genome-wide association study; OPC, oligodendrocyte.

and regulatory element regions of the *CLIC4* gene (Figure 3A). This suggests that rs12063304 is likely located within a regulatory region that influences *CLIC4* expression.

Next, we investigated the cell type-specific expression patterns using the SEA-AD dataset²⁵ of *CLIC4*. Expression levels of *CLIC4* were notably elevated in non-neuronal cell types, particularly in endothelial cells (Figure 3B). Additionally, in astrocytes, *CLIC4* was differentially expressed according to the final diagnosis of AD using the ROSMAP differential expression dataset (Figure 3C, Table S1 in supporting information).¹⁴ Individuals with a final cognitive diagnosis exhibited lower *CLIC4* expression in astrocytes. Moreover, in oligodendrocytes, significantly differential expression was observed in relation to the rate of change in memory and cognitive functions, with individuals exhibiting a higher rate of change showing higher *CLIC4* expression.

3.3 | Gene interactions and pathway overlaps between European and East Asian GWAS

We conducted a comparative analysis to investigate how genes identified in East Asian and European GWAS studies for AD are interconnected. We defined genes identified in East Asian GWAS ($n = 32$) comprising 28 genes reported in previous East Asian GWAS studies and four newly identified genes (*KCNJ6*, *MAPK1IP1L*, *AC016292.1*, and *AC100800.1*) through cell type-specific eQTLs and peak-to-gene links. Genes identified in European GWAS ($n = 76$) were identified as the nearest genes in large-scale studies. Notably, *SORL1*, *TREM2*, and *SHAPIN* were found in both East Asian and European populations.

Next, we conducted a network analysis incorporating genes from both East Asian and European GWAS (Figure 4A). We subsequently performed a pathway enrichment analysis to pinpoint pathways associated with interconnected genes. Clusters associated with the A β metabolic process, GABAergic synapse, and necroptosis pathways contained genes uniquely identified in East Asian GWAS. Specifically, the GABAergic synapse pathway included *KCNJ6*, newly identified through cell type-specific eQTL analysis. This cluster included *TSP0AP1* from European GWAS and *CACNA1A*, *KCNJ15* from East Asian GWAS. The

A β metabolic process cluster also included four genes identified in East Asian GWAS (*PTPRN2*, *SCARB2*, *APOC1*, and *SORCS1*). These findings highlight the interconnected nature of genes identified in East Asian and European GWAS within shared biological networks.

Finally, to explore the enrichment patterns of the global AD pathology burden, we performed GSEA using cell type-specific DEGs. We observed significant enrichment among DEG in astrocytes from both East Asian and European GWAS (Figure 4B). However, genes identified in East Asian GWAS showed enrichment in genes with negative correlations, whereas those from European GWAS exhibited enrichment in genes with positive correlations (Figure 4C). This highlights significant enrichment of genes from each population within the same cell type, yet with distinct enrichment patterns.

4 | DISCUSSION

In this study, we conducted a cell type-specific analysis to elucidate genes associated with AD risk loci identified in East Asian GWAS. By using cell type-specific eQTL data, we identified significant eQTLs for *MAPK1IP1L* and *KCNJ6*, which had not been previously reported in prior East Asian GWAS studies. Notably, *KCNJ6* interacted with *CACNA1A* and *KCNJ15* previously identified in East Asian GWAS. Cell type-specific peak-to-gene link analysis further revealed that the rs12063304 variant is mapped to a peak linked with four genes. Notably, the *CLIC4* gene was linked across six different cell types. Additionally, we observed that genes identified in European GWAS studies interact with those found in East Asian GWAS, and these genes showed significant enrichment in DEGs in astrocytes.

These novel insights into *KCNJ6* and *MAPK1IP1L* demonstrate the importance of cell type-specific eQTL analysis in uncovering previously unrecognized genetic interactions. Notably, *KCNJ6* is located within the Down syndrome critical region and has been implicated in Keppen–Lubinsky Syndrome (KPLBS), a rare disorder characterized by severe developmental delay and intellectual disability.³² Mutations in *KCNJ6* severely disrupt potassium channel function in KPLBS patients and significantly alter single-cell transcriptome profiles, particularly

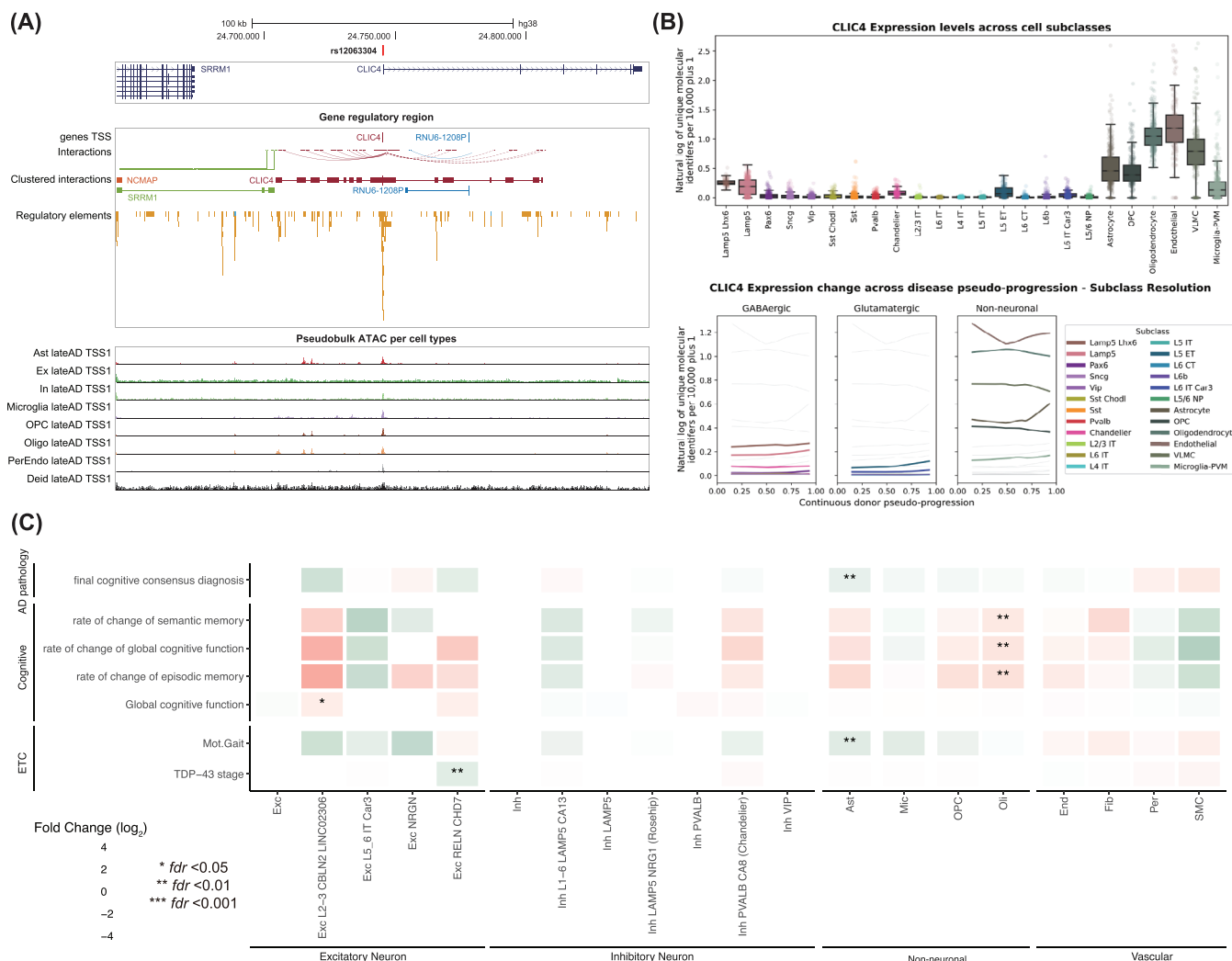


FIGURE 3 Regulatory region and cell type-specific expression analysis of *CLIC4*. (A) Visualization of the *CLIC4* gene region including gene regulatory regions and cell type-specific ATAC peaks. (B) Cell type-specific expression and expression patterns of *CLIC4* across pseudo-progression of AD pathology. Plotting was performed using SEA-AD gene expression trajectories viewer. (C) Heatmap of cell type-specific DEGs for *CLIC4*. Significance levels are indicated as * $fdr < 0.05$, ** $fdr < 0.01$, and *** $fdr < 0.001$. AD, Alzheimer's disease; Ast, astrocyte; END, endothelial cells; Exc, excitatory neuron; Fib, fibroblasts; Inh, inhibitory neuron; LB/TDP, Lewy Body/TAR DNA-binding protein; Mic, microglia; Oli, oligodendrocytes; OPC, oligodendrocyte progenitor cells; Per, pericytes; SEA-AD, Seattle Alzheimer's Disease Brain Cell Atlas; SMC, smooth muscle cells; Vasc, vascular cells.

impacting synaptic transmission and cell projection morphogenesis.³³ Moreover, evidence from AD mouse models suggests that *KCNJ6* may contribute to cognitive impairment.³⁴ These findings indicate that *KCNJ6* could play a crucial role in cognitive processes and may be a factor in the pathogenesis of AD.

Moreover, our results indicate that genes identified in East Asian GWAS studies, which were not previously recognized in European GWAS, interact with those reported in European studies. GSEA revealed that genes identified in both East Asian and European GWAS studies were enriched among the DEGs related to AD pathology in astrocytes. These findings emphasize the value of expanding genetic research to include multiple ethnic groups, which enhances our understanding of disease mechanisms and facilitates the identification of potential new therapeutic targets.

Several limitations should be considered. First, the sample sizes in East Asian GWAS are smaller compared to European cohorts, which may limit the statistical power.⁷ In South Korean studies, the issue of limited sample sizes is especially pronounced, underscoring the need for larger-scale GWAS to enhance the statistical power and interpretability of the findings. Additionally, this study used eQTL and peak-to-gene links from European snRNA and snATAC datasets. Gene expression patterns and eQTLs differ across populations, and thus using functional data from different populations might have limited our discovery. While the incorporation of East Asian eQTLs is essential to investigate population-specific variations, there is a deficiency of large-scale, cell type-specific eQTL and epigenome datasets derived from the East Asian cohorts. This limitation has led to Asian GWAS studies relying primarily on eQTL datasets derived from European

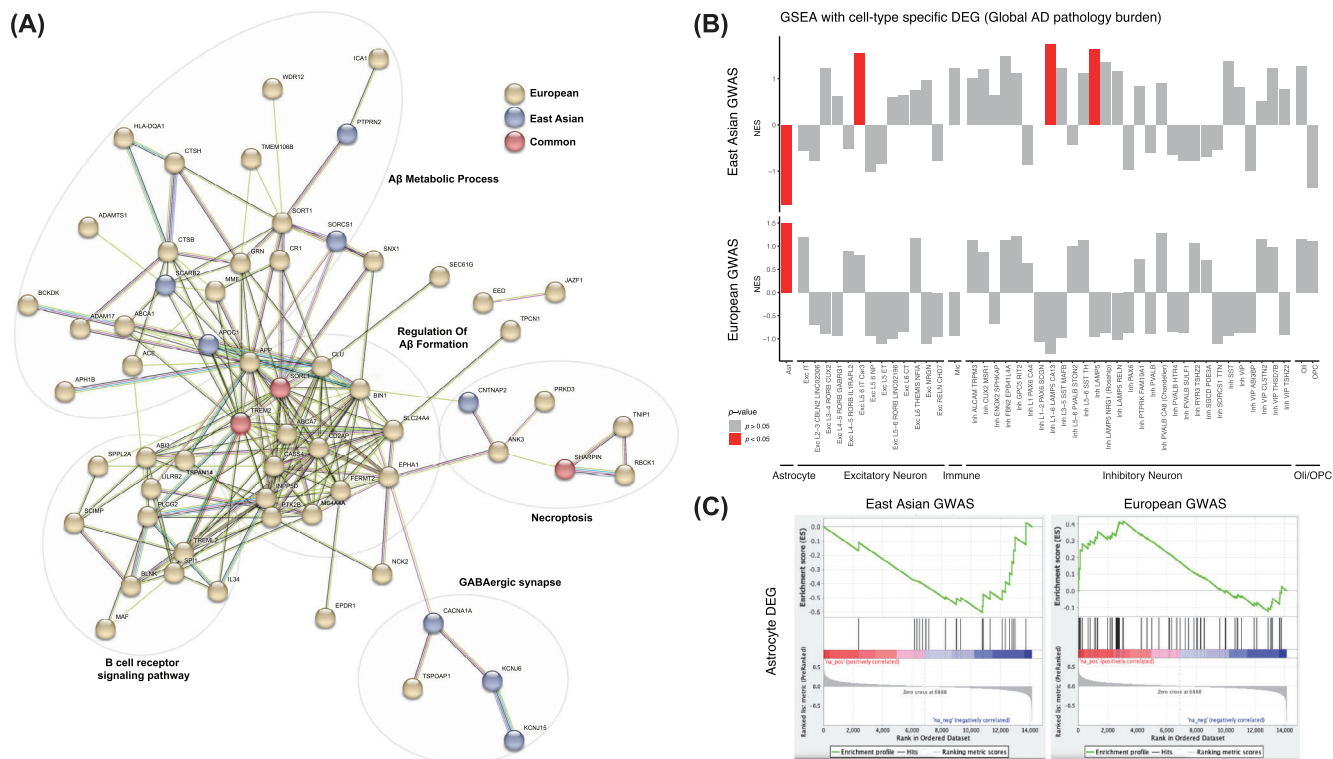


FIGURE 4 Network and gene set enrichment analysis of genes identified in East Asian and European GWAS. A, Network analysis of genes identified with East Asian and European GWAS. Yellow nodes represent European GWAS genes, blue nodes represent East Asian GWAS genes, and red nodes represent commonly identified genes. The different line colors represent interaction types: light blue for curated databases, pink for experimentally determined interactions, green for gene neighborhood, red for gene fusions, blue for gene co-occurrence, yellow-green for text mining, black for co-expression, and purple for protein homology. B, GSEA using cell type-specific DEGs of global AD pathology burden. The upper panel shows gene sets identified in East Asian GWAS, while the bottom panel displays those from European GWAS. Significant results are indicated by red bars ($p < 0.05$). C, GSEA analysis using astrocyte-specific DEGs of global AD pathology burden. Ast, astrocyte; Aβ, amyloid beta; DEG, differentially expressed genes; Exc, excitatory neuron; GSEA, gene set enrichment analysis; GWAS, genome-wide association study; Inh, inhibitory neuron; Mic, microglia; Oli, oligodendrocytes; OPC, oligodendrocyte progenitor cells.

populations.^{35,36} Therefore, comprehensive transcriptomic and epigenomic profiling data in East Asian populations are required to identify more functional AD loci for East Asians.

In summary, our study highlights the value of cell type-specific analyses in identifying AD risk genes from East Asian GWAS. These findings underscore the importance of incorporating diverse populations into genetic research to reveal new genetic factors and interactions relevant to AD. Addressing limitations such as smaller sample sizes and the need for comprehensive transcriptomic and epigenomic profiling in East Asian populations will be crucial for advancing our understanding of AD.

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CONFLICT OF INTEREST STATEMENT

Saykin receives support from multiple NIH grants (P30 AG010133, P30 AG072976, R01 AG019771, R01 AG057739, U19 AG024904, R01 LM013463, R01 AG068193, T32 AG071444, U01 AG068057, U01 AG072177, and U19 AG074879). He has also received support from Avid Radiopharmaceuticals, a subsidiary of Eli Lilly (in kind contribution of PET tracer precursor) and participated in Scientific Advisory Boards (Bayer Oncology, Eisai, Novo Nordisk, and Siemens

Medical Solutions USA, Inc) and an observational study monitoring board (MESA, NIH NHLBI), as well as external advisory committees for multiple NIA grants. He also serves as editor-in-chief of *Brain Imaging and Behavior*, a Springer-Nature Journal. Min Young Cho, Soumilee Chaudhuri, Shiwei Liu, Tamina Park, Yen-Ning Huang, Thea Rosewood, Paula J. Bice, Hong-Hee Won, and Kwangsik Nho have no competing interests to declare. The funders had no role in the study's design; the collection, analyses, or interpretation of data; the writing of the manuscript; or the decision to publish the results. Author disclosures are available in the [supporting information](#).

DATA AVAILABILITY STATEMENT

Cell-type specific eQTLs are available online in ROSMAP snRNA seq eQTL website (https://vmenon.shinyapps.io/rosmap_snRNAseq_eQTL/). All single cell-level analysis results from ROSMAP-MIT, including the DEGs, are available online in the AD/Aging Brain Atlas (https://compbio.mit.edu/ad_aging_brain/). The results of SEA-AD analysis are available online in the Allen Brain Map (<https://portal.brain-map.org/explore/seattle-alzheimers-disease>).

CONSENT STATEMENT

All participants provided informed consent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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